



A multifaceted pharmacist intervention to improve antihypertensive adherence: a cluster-randomized, controlled trial (HAPPy trial)

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SUMMARY

What is known and objectives: About half of all patients taking antihypertensives discontinue treatment by 12 months. There is potential for substantial health gains at both individual and population levels through improved treatment adherence. The objective was to evaluate a community pharmacist intervention to improve adherence with antihypertensive medicines with a view to improving blood pressure (BP) control.

Methods: Design: prospective, non-blinded, cluster-randomized, controlled trial.

Participants: adults with primary hypertension who obtained antihypertensives in the previous 6 months. Patients with poor refill adherence were preferentially identified with the help of a purpose-built software application.

Intervention: package comprising BP monitor; training on BP self-monitoring; motivational interviewing; medication use review; prescription refill reminders.

Follow-up: six months.

Primary outcome: change in proportion self-reporting medication adherence.

Secondary outcome: BP changes.

Results: Participants ($n = 395$; intervention – 207; control – 188) had a mean age of 66.7 years; 51.1% were males. The proportion of adherent participants increased in both groups but was not significantly different between groups [57.2% to 63.6% (control) vs. 60.0% to 73.5% (intervention), $P = 0.23$]. The mean reduction in systolic BP was significantly greater in the intervention group (10.0 mmHg vs. 4.6 mmHg; $P = 0.05$). The proportion of patients who were non-adherent at baseline and adherent at 6 months was 22.6% (95%CI 5.1–40.0%) higher in the intervention group (61.8% vs. 39.2%, $P = 0.007$). Among participants with baseline BP above target, reduction of systolic BP was significantly greater in the intervention group [by 7.2 mmHg (95%CI 1.6–12.8 mmHg); ($P = 0.01$)]. Among participants non-adherent at baseline and above target BP, the proportion reporting

adherence at 6 months was significantly greater in the intervention group [56.8% vs. 35.9%, $P = 0.039$].

What is new and conclusion: This community pharmacist intervention resulted in improved adherence to antihypertensive medication and reduced systolic BP.

WHAT IS KNOWN AND OBJECTIVES

Hypertension is the second leading cause of death globally and affects about one-third of the adult population in developed countries.¹ Appropriate antihypertensive therapy can improve cardiovascular outcomes, but patients need to use antihypertensive medicines consistently to optimize health benefits.² Adherent patients are significantly less likely to experience a first cardiovascular event, fatal other cardiovascular event or heart failure onset.³ Given that about half of all treated hypertensive patients discontinue treatment by 12 months,⁴ there is potential for substantial health gains at both individual and population levels through improved adherence.

A Cochrane review of randomized controlled trials (RCTs) of interventions to help patients follow prescriptions for medications for medical problems concluded that current approaches to improving adherence for chronic health problems are mostly complex and not very effective.⁵ While the blood pressure (BP)-lowering effects of pharmacist involvement with BP management are well defined via meta-analyses,^{6–9} uncertainty remains about the effect of pharmacist interventions on antihypertensive medication adherence. Methodological issues in existing trials that contribute to this uncertainty include high levels of baseline adherence allowing little room for improvement; large improvements to control group adherence; non-report of intervention or control group adherence at baseline or follow-up; sample size powered to detect significant changes to BP but not adherence; and use of standardized rather than individualized adherence interventions.

Robust evidence from RCTs regarding the efficacy of a well-defined, patient-tailored intervention by pharmacists to improve adherence to antihypertensive therapy is needed. This would confirm the usefulness of engaging pharmacists in the management of high BP in patients with adherence issues. It would also help to define a process for delivery of patient-specific adherence interventions. The aim of this study was to assess the efficacy of a

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well-defined, multifaceted, community pharmacist-delivered intervention to improve adherence to antihypertensive therapy in patients suspected of being non-adherent.

METHODS

A detailed study protocol for the Hypertension Adherence Program in Pharmacy (HAPPy) trial, conducted from July 2009 to January 2010, has been described previously.¹⁰ Briefly:

Pharmacy recruitment and training

Pharmacies from metropolitan, regional and remote areas in three Australian states (Victoria, Western Australia and Tasmania) were contacted by telephone and informed about the project.

Pharmacies were eligible to participate in the trial if they met all the following criteria:

- Used the pharmacy dispensing program 'FRED Dispense'® (Fred IT Group, Melbourne, Vic., Australia);
- Agreed to install a software application (MEDEMINECVD; University of Tasmania, Hobart, Tas., Australia) to identify patients who were using or had used antihypertensive medicines in the previous 6 months;
- Had at least 40 patients who were using or had used antihypertensive medicines in the previous 6 months;
- Had a private counselling area within the pharmacy;
- Were willing and able to undertake project training either face-to-face or online;
- Had time within the working week to allocate about one uninterrupted hour per patient to collect the baseline data; and
- Were able to follow up participants for at least 6 months from baseline.

Cluster randomization

In each state, pharmacies were stratified according to their location (metropolitan, regional and remote areas) and then randomized within strata to either Pharmacist Care Group (PCG) [intervention group] or Usual Care Group (UCG) [control group]. The pharmacies were stratified across the three states according to PhARIA, a pharmacy-specific measure of remoteness.¹¹ Randomization was performed at the pharmacy level to avoid any contamination likely to result from the same pharmacy recruiting and following up both intervention and control group patients. The randomization process was carried out by one of the investigators using the 'sealed envelope technique'. Participating pharmacists underwent training and subsequent assessment of knowledge and skills. Pharmacists were required to demonstrate a defined standard of competence. PCG pharmacists received training modules before commencement of the study, while UCG pharmacists received them after completion of data collection. To compensate for the pharmacist's time and costs in setting up a suitable environment for consultation, each participating pharmacy received AU\$500.

Patient recruitment

A data-mining software application, MEDEMINECVD, was designed specifically to be installed into the dispensing program common to all participating pharmacies (FRED Dispense®) to preferentially identify patients who had not collected their medication at regular intervals compatible with good adherence. The algorithm identified persons taking antihypertensive agents who had not had a refill dispensed for 45 days or longer at any time over the previous

6 months. (In Australia, the maximum quantity allowed to be dispensed under the Pharmaceutical Benefits Scheme for the selected antihypertensive agents is 30 days supply). The algorithm was used only as a screening process to identify suitable potential participants for the study. Expression of interest (EOI) letters were sent to potential participants to complete and return to the pharmacy.

Inclusion criteria

- Using, or having used in the previous 6 months, at least one antihypertensive medication belonging to the four common classes of antihypertensives in Australia¹² – angiotensin-converting enzyme inhibitors, angiotensin-II receptor antagonists, calcium channel blockers and beta-blockers – or fixed combinations of these antihypertensive medications with other antihypertensives (e.g. diuretics);
- A diagnosis of primary hypertension confirmed by patient-nominated general practitioner (GP);
- Aged 18 years or above; and
- Available for follow-up for at least 6 months from baseline.

Exclusion criteria

- Participation in other adherence promotion programs;
- Having had a pharmacist-conducted medicine review in the previous 12 months;
- Unable to communicate in English; and
- Not self-administering antihypertensive medicines.

Sample size

Around 50% of patients initiating an antihypertensive medication discontinue their medication within 12 months.⁴ To detect an improvement in adherence from 50% to 65% with 80% power and a two-sided *P*-value of 0.05, 182 patients were required per study group. To allow for potential dropouts (approximately 25% over 6 months), the intent was to recruit 225 patients per group, that is, a total of 450 patients at baseline. Based on the intracluster correlation coefficients (ICC) reported for demographic and clinical variables in primary care research studies in Australia¹³ and overseas,¹⁴ no inflation for cluster randomization was included as the ICC for change in adherence was expected to be negligible. Furthermore, with a minimum of 182 patients per group, this study had a >95% power to detect a change in BP equivalent to 50% of one standard deviation with a two-sided *P*-value of 0.05. A difference of this magnitude was perceived to be of clinical importance. This calculation was based on an average cluster size of eight and conservatively allows for an ICC to be as high as 0.10.

Data collection

Prior to baseline visits, individual GPs nominated by the participants were sent more detailed project information, including a request for confirmation of participating patients' hypertension diagnosis and their target BP.

Baseline visit

During the baseline visit to the pharmacy, the pharmacist measured the patient's BP using a digital BP monitor (Omron®HEM-790IT; Omron Healthcare Co Ltd, Muko, Kyoto, Japan) – average of two readings, as per WHO MONICA protocol.¹⁵ If

these differed by more than 10 mmHg for systolic or 5 mmHg for diastolic, a third reading was taken and the average of the two closest readings recorded. Adherence was measured using the Morisky scale¹⁶ and the Tool for Adherence Behaviour Screening (TABS).¹⁷ UCG participants continued to receive routine care. PCG participants received a package of interventions from the pharmacist for enhancing their antihypertensive medication adherence, which included the following:

- A home BP monitor (Omron®HEM-790IT) with the capacity to store and download BP readings to be used for discussion at three- and 6-month follow-ups;
- Training by the pharmacist on self-monitoring of BP;
- Motivational interviewing and education by the pharmacist to help patients improve their medication adherence and achieve target BP;
- Pharmacy-based medicines review to identify and resolve, where necessary, possible medication-induced hypertension (e.g. due to non-steroidal anti-inflammatory drugs, cold preparations, complementary medicines);
- Pharmacist-initiated dose administration aid (DAA), home-based medicines review and/or patient medication list, where necessary;
- Referral to a GP at the pharmacist's discretion; and
- Refill reminders (by SMS, telephone or mail), if they so chose, from their pharmacist 3 days before their antihypertensive medication was due to run out.

Interim visit

Pharmacist Care Group participants returned to the pharmacy at 3 months for review and action based on the downloaded home BP measures.

Final visit

Measures performed at baseline were repeated for both UCG and PCG patients at their 6-month visit to the pharmacy. Following final measurements, each UCG participant was eligible to receive the same package of interventions offered to the PCG at baseline.

Outcome measures

The primary outcome was change in the proportion of patients self-reporting adherence on the Morisky scale at the end of 6 months. Patient non-adherence was defined by a Morisky score of greater than zero. Secondary outcomes were changes in patients' systolic and/or diastolic BP and the proportion self-reporting adherence on the TABS. A differential of 15 or more between the TABS subscales (adherence and non-adherence) was regarded as good adherence.

Subgroup analysis

As improvements in the outcomes were dependent upon each patient's baseline levels, to target populations of specific interest, exploratory analyses were planned for the following subgroups:

- Participants who self-reported non-adherence (Morisky score >0);
- Participants with a baseline systolic and/or diastolic BP above the GP-nominated target; and
- Participants who had both self-reported non-adherence and BP above target.

Statistical analysis

Analysis was performed under the intention-to-treat principle using SAS version 9.2 (SAS Institute Inc., Cary, NC, USA).

Baseline characteristics between groups were compared using chi-square tests for equal proportion and reported as numbers (percentages). Continuous normally distributed variables were compared using Student *t*-tests and reported as means (standard deviations) whereas non-normally distributed variables were compared using Wilcoxon rank sum tests and reported as medians (interquartile range). To control for the effects of cluster randomization, group changes were compared using mixed linear and nonlinear modelling for normally and non-normally distributed data, respectively, with individual pharmacies treated as a random effect.¹⁸ Differences from cluster-adjusted models have been reported with 95% confidence intervals. A two-sided *P*-value ≤ 0.05 was considered to be statistically significant.

Ethics approval

This project was approved by the human research ethics committees of all participating universities. All pharmacists and patient participants provided written informed consent at the time of enrolment. The researchers had access only to de-identified patient data.

RESULTS

A total of 55 pharmacies were recruited and randomized to the PCG ($n = 29$) or the UCG ($n = 26$), in which 207 and 188 participants were recruited, respectively (Fig. 1). Baseline characteristics of the study participants are shown in Table 1. There were no significant differences between groups.

Of the 354 participants available for follow-up at 6 months, 343 (173 UCG and 170 PCG) had Morisky scores reported at both baseline and 6 months. The proportions adherent (Morisky score = 0) increased in both groups, PCG from 60% to 73.5% ($P = 0.003$) and UCG from 57.2% to 63.6% ($P = 0.1$). Although a significantly higher proportion of the PCG was adherent at 6 months, no between-group difference was detected for the change in proportion adherent [7.1% (95%CI -4.6 to 18.9%); $P = 0.23$]. Between-group changes in TABS differential score over the 6-month period did not reach statistical significance [0.1 (0.9–1.1); $P = 0.81$]. Proportions judged to be adherent using the TABS differential score also remained relatively consistent (<5% change) from baseline to 6 months within both groups. Both systolic and diastolic BP decreased significantly over the 6-month study period in both groups (Table 2). The reduction in systolic BP was significantly greater in the PCG than the UCG ($P = 0.05$), a difference of 5.3 mmHg (0.0–10.6 mmHg) between groups. The difference in diastolic BP between the PCG and UCG failed to reach statistical significance.

Subgroup analysis I: participants non-adherent at baseline

Subgroup analyses were performed on participants who self-reported non-adherence on the Morisky scale (score >0) at baseline and also had Morisky scores recorded at 6 months (UCG = 68; PCG = 75). The proportion of patients who became adherent was 22.6% (5.1–40.0%) higher in the PCG (61.8% vs. 39.2%; $P = 0.007$).

Both systolic and diastolic BP decreased significantly over the study period within both the UCG and the PCG (Table 3). The average change in systolic BP was statistically greater in PCG participants than those in the UCG [9.4% (2.2–16.6%); ($P = 0.01$)].

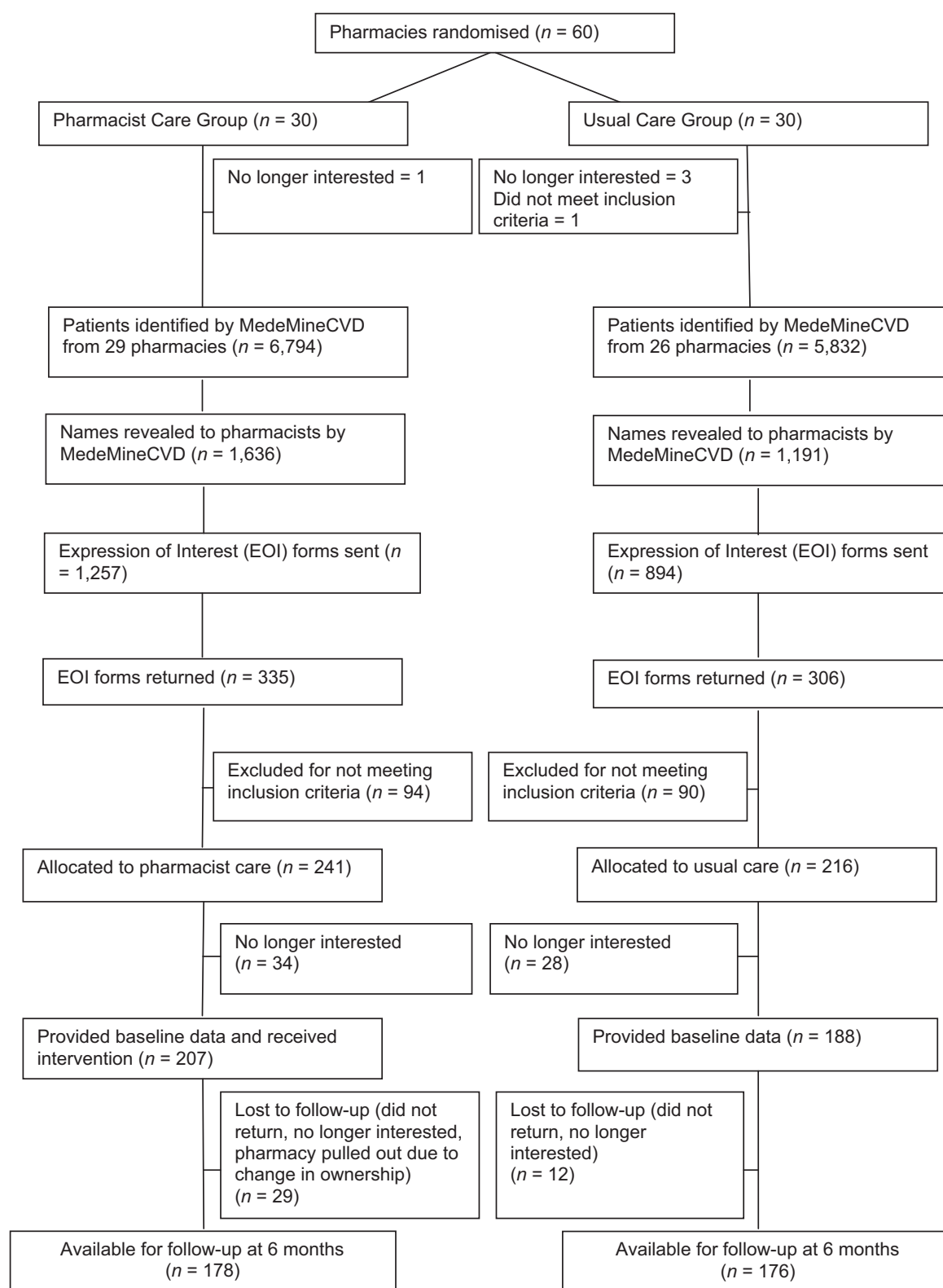


Fig. 1. Participant flow diagram.

Table 1. Baseline characteristics of participants

Characteristic	UCG (<i>n</i> = 188)	PCG (<i>n</i> = 207)
Male, <i>n</i> (%)	103 (54.8)	99 (47.8)
Mean age, years (SD)	66.6 (11.7)	66.8 (12.1)
Married or cohabiting, <i>n</i> (%)	130 (69.1)	137 (66.2)
Born in Australia, <i>n</i> (%)	140 (74.5)	141 (68.1)
Post-secondary education, <i>n</i> (%)	72 (38.3)	61 (29.5)
Annual household income <AUD75 000, <i>n</i> (%)	139 (84.8)	154 (79.4)
Concessional beneficiaries, ^a <i>n</i> (%)	129 (69.0)	151 (72.9)
Comorbidity		
Cardiovascular disease, <i>n</i> (%)	74 (39.4)	73 (35.3)
Diabetes, <i>n</i> (%)	31 (16.5)	40 (19.3)
Depression, <i>n</i> (%)	34 (18.1)	35 (16.9)
Median Charlson's comorbidity index score (IQR)	2 (1–3)	2 (1–3)
Non-adherence		
Morisky score >0, <i>n</i> (%)	79 (42.5)	85 (43.1)
Medication possession ratio (mean $\pm$ SD)	0.81 $\pm$ 0.25	0.81 $\pm$ 0.24
Blood pressure		
Mean systolic, mmHg (SD)	140.1 (20.2)	141.9 (20.0)
Mean diastolic, mmHg (SD)	83.2 (11.6)	84.3 (11.0)

AUD, Australian dollar; IQR, interquartile range; SD, standard deviation; PCG, Pharmacist Care Group; UCG, Usual Care Group.

^aEligible to receive government subsidized prescription medicines at a cheaper rate.

Table 2. Changes in BP between baseline and 6 months

BP reading	Baseline	6 months	Reduction	<i>P</i>
Systolic BP				
UCG (<i>n</i> = 176)	140.1 $\pm$ 22.5	135.3 $\pm$ 22.3	4.6 $\pm$ 25.3	0.01
PCG (<i>n</i> = 176)	141.9 $\pm$ 22.4	131.7 $\pm$ 22.0	10.0 $\pm$ 25.0	<0.001
Difference between UCG and PCG (95% CI)			5.3 (0.0–10.6)	0.05
Diastolic BP				
UCG (<i>n</i> = 176)	83.2 $\pm$ 14.5	78.8 $\pm$ 13.8	4.2 $\pm$ 14.7	<0.001
PCG (<i>n</i> = 176)	84.3 $\pm$ 14.4	80.2 $\pm$ 13.6	4.5 $\pm$ 14.5	<0.001
Difference between UCG and PCG (95% CI)			0.3 (–2.8 to 3.4)	0.85

BP, blood pressure; PCG, Pharmacist Care Group; UCG, Usual Care Group.

Subgroup analysis II: participants above BP target at baseline

In participants with baseline BP (systolic and/or diastolic) above the GP-nominated target (UCG = 122; PCG = 113), within-groups reductions in systolic and diastolic BP over the study period were significant in both groups (Table 4). The systolic BP reduction was significantly greater for PCG participants than UCG participants [7.2 mmHg (95%CI 1.6–12.8 mmHg); *P* = 0.01].

Table 3. Changes in BP in participants non-adherent at baseline

BP reading	Baseline	6 months	Reduction	<i>P</i>
Systolic BP				
UCG (<i>n</i> = 75)	138.2 $\pm$ 20.3	134.2 $\pm$ 19.8	3.6 $\pm$ 21.7	<0.001
PCG (<i>n</i> = 68)	141.8 $\pm$ 20.3	128.7 $\pm$ 19.3	13.0 $\pm$ 21.7	<0.001
Difference between UCG and PCG			9.4 (2.2–16.6)	0.01
Diastolic BP				
UCG (<i>n</i> = 75)	83.9 $\pm$ 12.9	80.5 $\pm$ 11.2	2.8 $\pm$ 12.5	0.05
PCG (<i>n</i> = 68)	84.9 $\pm$ 12.8	79.5 $\pm$ 11.0	6.0 $\pm$ 12.3	<0.001
Difference between UCG and PCG			3.2 (–0.8 to 7.2)	0.11

BP, blood pressure; PCG, Pharmacist Care Group; UCG, Usual Care Group.

Table 4. Changes in BP in participants above target BP at baseline

BP readings	Baseline	6 months	Reduction	<i>P</i>
Systolic BP				
UCG (<i>n</i> = 122)	147.9 $\pm$ 17.9	138.3 $\pm$ 22.6	–9.1 $\pm$ 21.2	<0.001
PCG (<i>n</i> = 113)	150.0 $\pm$ 17.9	133.4 $\pm$ 22.3	–16.3 $\pm$ 21.2	<0.001
Difference between UCG and PCG			7.2 (1.6–12.8)	0.01
Diastolic BP				
UCG (<i>n</i> = 122)	86.8 $\pm$ 10.6	80.0 $\pm$ 12.7	6.6 $\pm$ 12.6	<0.001
PCG (<i>n</i> = 113)	88.3 $\pm$ 10.6	81.2 $\pm$ 12.5	7.2 $\pm$ 12.3	<0.001
Difference between UCG and PCG			0.5 (–2.7 to 3.7)	0.76

BP, blood pressure; PCG, Pharmacist Care Group; UCG, Usual Care Group.

Subgroup analysis III: participants non-adherent and above BP target at baseline

Among participants self-reporting non-adherence at baseline (Morisky score >0) and with BP (systolic and/or diastolic) above the GP target (UCG = 52; PCG = 44), the proportion who became adherent at 6 months was significantly greater in the PCG [56.8% (*n* = 25) vs. 35.9% (*n* = 19), *P* = 0.039].

Both systolic and diastolic BP decreased significantly over the study period in both groups (Table 5). The difference of 13.2 mmHg (4.6–21.8 mmHg) in systolic BP between the PCG and UCG was statistically significant (*P* = 0.004).

The changes in systolic BP across the whole group and three subgroups are presented in Fig. 2.

DISCUSSION

The HAPPY intervention is among the most comprehensive interventions for antihypertensive adherence evaluated via RCT, particularly in primary care.^{5,19,20} This is the first major antihypertensive adherence trial that sought to preferentially identify non-adherent individuals for inclusion – previous large trials have overwhelmingly selected participants on the basis of uncontrolled

Table 5. Changes in BP in patients non-adherent and not achieving target BP at baseline

BP readings	Baseline	6 months	Reduction	P
Systolic BP				
UCG (<i>n</i> = 52)	145.5 ± 17.8	137.9 ± 17.4	7.1 ± 21.1	0.02
PCG (<i>n</i> = 44)	150.3 ± 17.8	129.9 ± 17.4	20.3 ± 21.1	<0.001
Difference between UCG and PCG			13.2 (4.6–21.8)	0.004
Diastolic BP				
UCG (<i>n</i> = 52)	87.5 ± 11.7	81.5 ± 10.7	5.2 ± 11.9	0.002
PCG (<i>n</i> = 44)	89.7 ± 11.7	81.1 ± 10.5	9.2 ± 11.7	<0.001
Difference between UCG and PCG			3.9 (–0.9 to 8.7)	0.11

BP, blood pressure; PCG, Pharmacist Care Group; UCG, Usual Care Group.

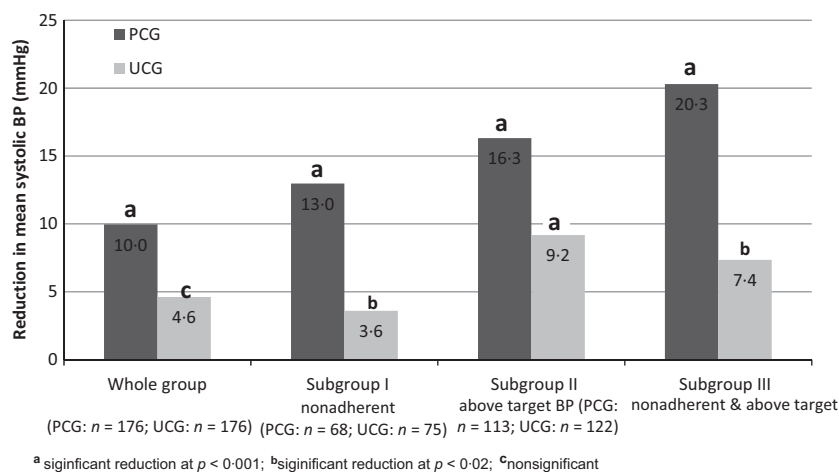
BP rather than medication non-adherence. Our adherence-focused pharmacist intervention tailored to individual patients, while not demonstrating significantly improved adherence on a number of measures (previously published)²¹ in the intervention group vs. the control group helped to improve the management of hypertension. The intervention was associated with a significantly greater reduction in mean systolic BP in comparison with usual care. Comparison with published meta-analyses of antihypertensive drugs trials suggests an overall antihypertensive effect comparable to adding an additional pharmacological agent.²² Subgroup analysis of non-adherent participants showed a decrease in systolic BP in the PCG of the order of 13 mmHg, which was significantly greater than the drop in the UCG by a margin of 9.4 mmHg. Using RCTs of drug therapy as a guide, this improvement could feasibly reduce the risk of major cardiovascular events for this subgroup by 20–30%.²² Participants who were above target BP at baseline also showed a significant decrease in systolic BP in the PCG (of the order of 16 mmHg). The subgroup analyses suggest that careful screening of patients on the basis of both self-reported adherence and BP control might help to maximize impact. These exploratory subgroup analyses suggest possibilities for future hypothesis testing.

Previous pharmacist intervention studies in the management of cardiovascular disease risk factors have focussed on educational interventions directed to patients, patient-reminder systems, medication management, feedback to healthcare professionals, measurement of CVD risk factors and/or review of laboratory data by pharmacists during follow-up, and educational interventions directed to healthcare professionals.⁷ A recently published study by Svarstad *et al.*,²³ the TEAM trial, was conducted on a cohort of black patients with hypertension (*n* = 576) through community pharmacies (*n* = 28) belonging to several chains. The intervention involved team education, individualized counselling and adherence monitoring with up to six visits over a period of 6 months and follow-up at 12 months. Adherence interventions by other health professionals for patients with chronic conditions have tended to involve standardized interventions with a focus on cognitive or behavioural counselling and education, sometimes with extra features such as peer support, reminders or adherence monitoring.^{5,6} Comprehensive interventions, especially involving system-level intervention options, are uncommon. HAPPy included a patient management system, patient-reminder systems, motivational interviewing and BP self-monitoring and downloading of results for use as an educational and motivational tool.

In contrast to the HAPPy trial, the TEAM trial demonstrated significant and sustained improvements in adherence in the intervention group at 6 months.²³ The TEAM trial, however, comprised a more intensive intervention (up to six visits) than was considered feasible in the HAPPy trial, which was designed to be readily integrated into regular practice. In addition, refill data were likely to be more reliable in the TEAM trial as pharmacies within chains were electronically linked and patients were encouraged to stay with their particular chain.

The reduction in systolic BP achieved in the HAPPy trial aligns closely with the findings of a meta-analysis of 2619 patients in eight studies, which found that pharmacist interventions significantly reduced systolic BP from baseline to endpoint between intervention and control groups of -4.9 ± 0.9 mmHg ($P < 0.001$).⁹ Similar reductions in systolic BP were achieved at 6 months in both the HAPPy trial (5.3 mmHg) and the TEAM trial²³ (7.3 mmHg), with participants in the TEAM trial having higher baseline BP and more intensive intervention than those in the HAPPy trial.

Improvements in BP were also demonstrated for participants who did not self-report adherence problems in our study. This

**Fig. 2.** Reduction in systolic blood pressure over 6 months.

might reflect the generic benefits beyond adherence of some aspects of the intervention. Greater patient awareness about their BP may have driven improvements in lifestyle also. It is possible that some patients simply did not admit non-adherence, but still benefited from the adherence intervention. Regardless, it would appear that such an intervention should be considered for any patient with suboptimal BP control.

Due to chance alone, the number of potential participants identified by MEDEMINCVD was 17% higher for the PCG than for the UCG (6794 vs. 5832). Consequently, more names were revealed to pharmacists in the PCG (1636 vs. 1191) and a higher number of expressions of interest forms were sent to potential PCG participants (1257 vs. 894). While this may reflect a greater percentage of larger pharmacies in the UCG, there was no apparent relationship with change in either adherence or BP with pharmacy size. The retention rate in the UCG was higher than in the PCG (81% vs. 74% $P = 0.07$). This may be attributed to UCG participants having less treatment burden during the trial and an incentive to complete the study, as they would receive the intervention upon completion.

Although MEDEMINCVD was designed to identify patients apparently experiencing problems with adherence or persistence to their medications, high proportions reported good adherence at baseline. In the Australian setting, refill data may not be a reliable measure of adherence because data from different pharmacies are not linked and patients often go to multiple pharmacies for their refills. As the algorithm was used only as a screening tool, we accepted the fact that, while not being completely accurate, it would at least reduce the proportion of false positives (adherent people) selected. There is no gold standard for measuring adherence; self-reported measures are subject to the Hawthorne effect and to recall bias, whereas calculations from refill data are limited by incomplete information.²⁴ Estimation of adherence using medication refill data over a 6-month period may have been skewed as one supply missing from the pharmacy record could have made a difference to the estimate; 12 months data would have been preferable, but was not possible with the funding body's timelines.

Possible contributors to the failure to demonstrate a significant improvement in adherence have been discussed elsewhere²¹ and include the fact that the proposed magnitude of improvement in baseline adherence may have been ambitious, meaning that a larger sample size was needed to detect the actual differences in proportions adherent at baseline and 6 months. Also, sample sizes were not calculated to detect specific changes in the various adherence measures and may have been inadequate for that purpose. Differences in the results using different measures may reflect differences in sensitivity among the measures. This study has several other limitations, some of which are inherent in this type of research.

In the absence of a gold standard for measuring adherence, we chose self-report using the Morisky score in preference to refill data that were known to be unreliable as a measure. Self-reporting carries with it limitations such as poor participant recall, potential for the Hawthorne effect and possible low sensitivity to change. There was also a potential Hawthorne effect on prescribers that may have resulted in intensified prescribing; however, this was likely to be minimal, as the pharmacists contacted GPs prior to randomization only to establish participant eligibility, and routine contact was not part of the intervention. More importantly, in the context of a RCT, such effects would have been balanced across the study groups. Pharmacists in the intervention group may have been tempted to falsify BP results. Wherever possible, intervention studies should include longer-term follow-up to assess sustainability of any changes achieved, as the effect often decreases over time. Economic evaluations should be undertaken to determine the cost-effectiveness of the intervention and the potential sustainability of the service.

While the observed improvement in adherence within the PCG was similar to what was anticipated (13.5%), the net improvement in adherence was 7.1%, due to the UCG also displaying an increase in adherence (6.4%). The systematic review by Morgado *et al.*⁹ also found that pharmacist-delivered adherence interventions for antihypertensive therapy resulted in significantly reduced systolic and diastolic BP, but that the changes to adherence were uncertain. The magnitude of systolic BP lowering within the PCG was similar to that reported by Morgado *et al.*,⁹ although PCG participants received only one interim visit (at 3 months). Moreover, BP was significantly reduced for PCG patients who were above target at baseline when compared with their UCG counterparts. From a public health perspective, this is the more important outcome.

Several key components of the overall intervention are becoming increasingly common in pharmacy practice. SMS reminders, medicine use reviews and automated BP monitors are now commonplace in many countries;^{25–31} private counselling areas are increasingly a standard requirement for pharmacies in Australia and many other countries;^{32,33} and reimbursement systems for pharmacist interventions are in place in many developed countries making such interventions increasingly viable.^{32,34}

WHAT IS NEW AND CONCLUSION

This multifaceted community-based pharmacist intervention achieved a significant reduction in BP. It is noteworthy that the process was implemented successfully across 55 community pharmacies and that patient recruitment was feasible in this setting. Based on this evidence, community pharmacists should be actively engaged in providing tailored advice to appropriate patients with hypertension. The impact can be maximized by targeting those with both antihypertensive medication non-adherence and above target BP.

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